

Ex.3.10.1 : An axial load of 400 kN is applied at 20°C to the rod, as shown in Fig. P. 3.10.1(a). The temperature is then raised to 50°C. The material for element 1 is aluminium and for element 2 is steel. The relative properties for element 1 and 2 are as shown below.

Property	Element 1	Element 2
Material	Aluminium	Steel
Modulus of Elasticity	$70 \times 10^9 \text{ N/m}^2$	$200 \times 10^9 \text{ N/m}^2$
Cross-section Area	900 mm^2	1200 mm^2
Coefficient of Thermal Expansion	$2.3 \times 10^{-6} \text{ per } ^\circ\text{C}$	$11.7 \times 10^{-6} \text{ per } ^\circ\text{C}$

Determine :

- (i) the element stiffness matrix; (ii) the load vector;
 (iii) the nodal displacements; and (iv) the stresses in each element.

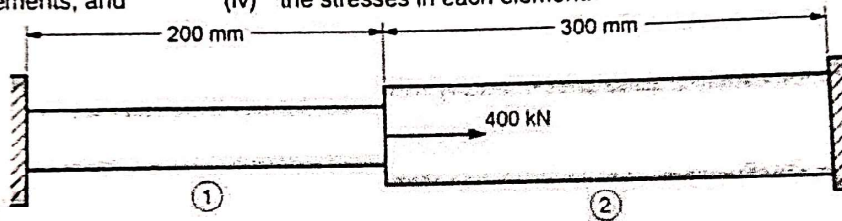


Fig. P. 3.10.1(a)

Soln. :

Given : $A_1 = 900 \text{ mm}^2$; $A_2 = 1200 \text{ mm}^2$; $l_1 = 200 \text{ mm}$;
 $l_2 = 300 \text{ mm}$; $E_1 = 70 \times 10^3 \text{ N/mm}^2$; $E_2 = 200 \times 10^3 \text{ N/mm}^2$;
 $\alpha_1 = 2.3 \times 10^{-6} \text{ per } ^\circ\text{C}$; $\alpha_2 = 11.7 \times 10^{-6} \text{ per } ^\circ\text{C}$; $\Delta T = 50 - 20 = 30^\circ\text{C}$;
 $P_2 = 400 \times 10^3 \text{ N}$.

1. Discretization of stepped bar :

As a load of 400 kN is acting at a distance of 200 mm from left support, the stepped bar is modeled with two elements, as shown in Fig. P. 3.10.1(b).

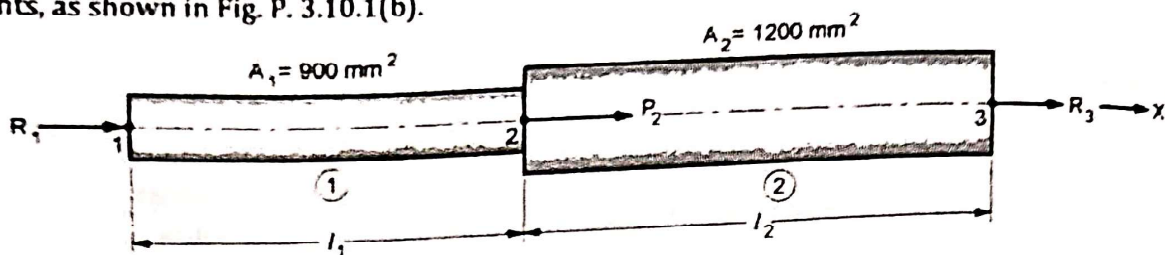


Fig. P. 3.10.1(b)

- The element connectivity for the assembly is given in Table P. 3.10.1.

Table P. 3.10.1 : Element Connectivity

Element Number ①	Global Node Number 'n' of	
	Local Node 1	Local Node 2
①	1	2
②	2	3

- Total d.o.f. of assembly, $N = \text{D.O.F. per node} \times \text{Number of nodes in assembly} = 1 \times 3 = 3$
- The dimensions of global stiffness matrix, $[K] = (3 \times 3)$
- The dimensions of global load vector, $\{F\} = (3 \times 1)$
- The dimensions of global nodal displacement vector, $\{U_N\} = (3 \times 1)$

2. Element stiffness matrices :

- Element 1 :

$$[k]_1 = \frac{A_1 E_1}{l_1} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = \frac{900 \times 70 \times 10^3}{200} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = 315 \times 10^3 \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \frac{1}{2} \text{ N/mm} \quad (a)$$

- Element 2 :

$$[k]_2 = \frac{A_2 E_2}{l_2} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = \frac{1200 \times 200 \times 10^3}{300} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = 315 \times 10^3 \begin{bmatrix} 2.54 & -2.54 \\ -2.54 & 2.54 \end{bmatrix} \frac{2}{3} \text{ N/mm} \quad (b)$$

3. Global stiffness matrix :

$$[K] = [k]_1 + [k]_2$$

The global stiffness matrix is obtained by assembling element stiffness matrices $[k]_1$ and $[k]_2$ such that the elements of each stiffness matrix are placed at the appropriate locations in the global stiffness matrix.

$$[K] = 315 \times 10^3 \begin{bmatrix} 1.0 & -1.0 & 0 \\ -1.0 & 1.0 + 2.54 & -2.54 \\ 0 & -2.54 & 2.54 \end{bmatrix} \frac{1}{2} \text{ N/mm} = 315 \times 10^3 \begin{bmatrix} 1.0 & -1.0 & 0 \\ -1.0 & 3.54 & -2.54 \\ 0 & -2.54 & 2.54 \end{bmatrix} \frac{1}{2} \text{ N/mm} \quad (c)$$

4. Element Load Vectors :

$$\{P_i\} = \begin{Bmatrix} R_1 \\ P_2 \\ R_3 \end{Bmatrix} \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} \text{ N} = \begin{Bmatrix} R_1 \\ 400 \times 10^3 \\ R_3 \end{Bmatrix} \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} \text{ N} \quad \dots(d)$$

• Temperature load vector for element 1 :

$$\{f\}_1 = A_1 E_1 \alpha_1 \Delta T \begin{Bmatrix} -1 \\ 1 \end{Bmatrix} = 900 \times 70 \times 10^3 \times 2.3 \times 10^{-6} \times 30 \begin{Bmatrix} -1 \\ 1 \end{Bmatrix} = \begin{Bmatrix} -4347 \\ 4347 \end{Bmatrix} \begin{matrix} 1 \\ 2 \end{matrix} \text{ N} \quad \dots(e)$$

• Temperature load vector for element 2 :

$$\{f\}_2 = A_2 E_2 \alpha_2 \Delta T \begin{Bmatrix} -1 \\ 1 \end{Bmatrix} = 1200 \times 200 \times 10^3 \times 11.7 \times 10^{-6} \times 30 \begin{Bmatrix} -1 \\ 1 \end{Bmatrix} = \begin{Bmatrix} -84240 \\ 84240 \end{Bmatrix} \begin{matrix} 2 \\ 3 \end{matrix} \text{ N} \quad \dots(f)$$

5. Global load vector :

$$\begin{aligned} \{F\} &= \{f_1\} + \{f_2\} + \{P_i\} \\ &= \begin{Bmatrix} -4347 \\ 4347 \\ 0 \end{Bmatrix} \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} + \begin{Bmatrix} 0 \\ -84240 \\ 84240 \end{Bmatrix} \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} + \begin{Bmatrix} R_1 \\ 400000 \\ R_3 \end{Bmatrix} \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} \\ \text{or } \{F\} &= \begin{Bmatrix} R_1 - 4347 \\ 320107 \\ R_3 + 84240 \end{Bmatrix} \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} \text{ N} \quad \dots(g) \end{aligned}$$

6. Global nodal displacement vector :

$$\{U_N\} = \begin{Bmatrix} U_1 \\ U_2 \\ U_3 \end{Bmatrix} \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} \text{ mm} \quad \dots(h)$$

7. Global stiffness-nodal displacement-load relationship :

Hence, the resultant matrix equation for the assembly is,

$$[K] \{U_N\} = \{F\}$$

$$315 \times 10^3 \begin{bmatrix} 1.0 & -1.0 & 0 \\ -1.0 & 3.54 & -2.54 \\ 0 & -2.54 & 2.54 \end{bmatrix} \begin{Bmatrix} U_1 \\ U_2 \\ U_3 \end{Bmatrix} = \begin{Bmatrix} R_1 - 4347 \\ 320107 \\ R_3 + 84240 \end{Bmatrix} \quad \dots(i)$$

8. Nodal displacements :

- At nodes 1 and 3, there are rigid supports. Hence $U_1 = 0$ and $U_3 = 0$
- As d.o.fs. 1 and 3 are fixed, using *elimination approach*, first and third rows and columns can be eliminated from Equation (i).

$$\text{Hence, } 315 \times 10^3 [3.54] [U_2] = \{320107\} \quad \dots(j)$$

$$\therefore 315 \times 10^3 \times 3.54 U_2 = 320107$$

4. Element Load Vectors :

$$\{P_i\} = \begin{Bmatrix} R_1 \\ P_2 \\ R_3 \end{Bmatrix} \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} N = \begin{Bmatrix} R_1 \\ 400 \times 10^3 \\ R_3 \end{Bmatrix} \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} N \quad \dots(d)$$

• Temperature load vector for element 1 :

$$\{f\}_1 = A_1 E_1 \alpha_1 \Delta T \begin{Bmatrix} -1 \\ 1 \end{Bmatrix} = 900 \times 70 \times 10^3 \times 2.3 \times 10^{-6} \times 30 \begin{Bmatrix} -1 \\ 1 \end{Bmatrix} = \begin{Bmatrix} -4347 \\ 4347 \end{Bmatrix} \begin{matrix} 1 \\ 2 \end{matrix} N \quad \dots(e)$$

• Temperature load vector for element 2 :

$$\{f\}_2 = A_2 E_2 \alpha_2 \Delta T \begin{Bmatrix} -1 \\ 1 \end{Bmatrix} = 1200 \times 200 \times 10^3 \times 11.7 \times 10^{-6} \times 30 \begin{Bmatrix} -1 \\ 1 \end{Bmatrix} = \begin{Bmatrix} -84240 \\ 84240 \end{Bmatrix} \begin{matrix} 2 \\ 3 \end{matrix} N \quad \dots(f)$$

5. Global load vector :

$$\begin{aligned} \{F\} &= \{f_1\} + \{f_2\} + \{P_i\} \\ &= \begin{Bmatrix} -4347 \\ 4347 \\ 0 \end{Bmatrix} \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} + \begin{Bmatrix} 0 \\ -84240 \\ 84240 \end{Bmatrix} \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} + \begin{Bmatrix} R_1 \\ 400000 \\ R_3 \end{Bmatrix} \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} \\ \text{or } \{F\} &= \begin{Bmatrix} R_1 - 4347 \\ 320107 \\ R_3 + 84240 \end{Bmatrix} \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} N \quad \dots(g) \end{aligned}$$

6. Global nodal displacement vector :

$$\{U_N\} = \begin{Bmatrix} U_1 \\ U_2 \\ U_3 \end{Bmatrix} \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} \text{ mm} \quad \dots(h)$$

7. Global stiffness-nodal displacement-load relationship :

Hence, the resultant matrix equation for the assembly is,

$$[K] \{U_N\} = \{F\}$$

$$315 \times 10^3 \begin{bmatrix} 1.0 & -1.0 & 0 \\ -1.0 & 3.54 & -2.54 \\ 0 & -2.54 & 2.54 \end{bmatrix} \begin{Bmatrix} U_1 \\ U_2 \\ U_3 \end{Bmatrix} = \begin{Bmatrix} R_1 - 4347 \\ 320107 \\ R_3 + 84240 \end{Bmatrix} \quad \dots(i)$$

8. Nodal displacements :

At nodes 1 and 3, there are rigid supports. Hence $U_1 = 0$ and $U_3 = 0$

As d.o.fs. 1 and 3 are fixed, using *elimination approach*, first and third rows and columns can be eliminated from Equation (i).

$$\text{Hence, } 315 \times 10^3 [3.54] [U_2] = \{320107\}$$

$$\therefore 315 \times 10^3 \times 3.54 U_2 = 320107 \quad \dots(j)$$

$$\therefore U_2 = 0.287 \text{ mm}$$

$$U_1 = 0; U_2 = 0.287 \text{ mm and } U_3 = 0$$

$$\{U_N\} = \begin{Bmatrix} 0 \\ 0.287 \\ 0 \end{Bmatrix} \text{ mm}$$

9. Support reactions :

From matrix Equation (I),

$$315 \times 10^3 \times (U_1 - U_2) = R_1 - 4347$$

$$315 \times 10^3 \times (-2.54 U_2 + 2.54 U_3) = R_3 + 84240$$

From Equation (k),

$$315 \times 10^3 \times (0 - 0.287) = R_1 - 4347$$

$$\therefore R_1 = -86058 \text{ N}$$

From Equation (l),

$$315 \times 10^3 \times (-2.54 \times 0.287 + 0) = R_3 + 84240$$

$$\therefore R_3 = -313868.7 \text{ N}$$

$$R_1 = -86.058 \text{ kN and } R_3 = -313.868 \text{ kN}$$

10. Stresses in elements :

Stress in any element is given by,

$$\sigma = E [B] \{u_N\} - E \alpha \Delta T = \frac{E}{l} [-1 \ 1] \{u_N\} - E \alpha \Delta T$$

• Element 1:

$$\begin{aligned} \sigma_1 &= \frac{E_1}{l_1} [-1 \ 1] \begin{Bmatrix} U_1 \\ U_2 \end{Bmatrix} - E_1 \alpha_1 \Delta T = \frac{E_1}{l_1} (-U_1 + U_2) - E_1 \alpha_1 \Delta T \\ &= \frac{70 \times 10^3}{200} (-0 + 0.287) - 70 \times 10^3 \times 2.3 \times 10^{-6} \times 30 \end{aligned}$$

$$\text{or } \sigma_1 = 95.62 \text{ N/mm}^2$$

• Element 2:

$$\begin{aligned} \sigma_2 &= \frac{E_2}{l_2} [-1 \ 1] \begin{Bmatrix} U_2 \\ U_3 \end{Bmatrix} - E_2 \alpha_2 \Delta T \\ &= \frac{E_2}{l_2} (-U_2 + U_3) - E_2 \alpha_2 \Delta T \\ &= \frac{200 \times 10^3}{300} (-0.287 + 0) - 200 \times 10^3 \times 11.7 \times 10^{-6} \times 30 \end{aligned}$$

$$\text{or } \sigma_2 = -261.5 \text{ N/mm}^2$$

Steps to solve the problem:

- Preprocessor → Element Type → Add/Edit/Delete → Link → 3D finite str 180
- Material Properties → Material Models → Structural → Linear → Elastic → Isotropic → EX → $70e3$ → PRXY → 0.3 → Thermal Expansion → Secant Coefficient → Isotropic → ALPX → $2.3e-6$ → OK
- Material → New Model → ID → 2 → OK → Structural → Linear → Elastic → Isotropic → EX → $200e3$ → PRXY → 0.3 → Thermal Expansion → Secant Coefficient → Isotropic → ALPX → $11.7e-6$ → OK → Exit
- Sections → Link → Add → ID → 1 → Name → 1 → Area → 900 → Apply → ID → 2 → Name → 2 → Area → 1200 → OK
- Modeling → Create → Keypoints → In Active CS → 1 → 0, 0, 0 → Apply → 2 → 200, 0, 0 → Apply → 3 → 500, 0, 0 → OK
- Create Lines → Lines → Straight Lines → Select Keypoints 1 & 2 and 2 & 3
- Meshing → Mesh Attributes → Picked Lines → Select Line 1 → OK → Select Material number 1 → Select Element Type Number 1 LINK 180 → Select Section 1 → Apply → Select Line 2 → OK → Select Material number 2 → Select Element Type Number 1 LINK 180 → Select Section 2 → OK
- Size Cntrl → Manual Size → Lines → Picked Lines → Pick All → NDIV → 1 → OK
- Mesh → Lines → Pick All
- Loads → Define Loads → Apply → Structural → Displacements → On Keypoints → Select Keypoint 1 and 3 → OK → All DOF → 0 → OK
- Force/Moments → On Keypoints → Select Keypoint 2 → OK → FX → $4e5$ → OK
- Temperature → On Lines → Select both lines → OK → Temperature → 30 → OK
- Solution → Solve → Current LS → OK
- General Postproc → List Results → Reaction Solu → All Items → OK
- Nodal Solution → DOF Solution → Displacement Vector Sum
- Element Table → Define Table → ADD → By Sequence Num → SMISC, 1 → Apply → ADD → By Sequence Num → LS, 1 → OK
- List Results → Element Table Data → Select SMISC1 and LS1 → OK

Conclusion:

Solution obtained for the stepped bar can be validated using ANSYS as follows

	Nodal Displacement (mm)			Element Stress (N/mm ²)		Reaction Force, (N)	
	Node 1	Node 2	Node 3	Element 1	Element 2	Node 1	Node 3
Finite Element Method (Manual Calculation)							
Finite Element Method (ANSYS)							